Predicting protein functions with deep learning and multi-source data

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Introduction

- Expert curation is critical for protein function annotation, but it cannot cope with the exponential growth of protein sequences data [1].
- Extracting features suitable for protein function prediction requires exhaustive multi-source data preprocessing [2].
- Methods for automatic function prediction are being constantly developed, but their performance is still subject to improvement [2-3].

Multi-source data

- 5,799 *S. Cerevisiae* proteins.
- Protein association matrices for model input:
  - *PSD:* Proteins sequence edit distance;
  - *TED*: Correlation of microarray expression data (GSE95152);
  - *GOCo*: Co-occurrence in GO sub-ontologies;
  - *ReactomeCo*: Co-occurrence in Reactome pathways (1,459 proteins).
- GO annotations: BP (3,360 proteins), CC (4,200 proteins), MF (2,613 proteins).

Proposal: Deep learning model

- Input data from association matrices defined between proteins.
- Fully connected layers, with batch normalization and dropout (**Figure 1**).
- Protein annotations modeled as one-hot encoded vectors.
- Model output taken from four consecutive layers modeling the hierarchical levels of the GO.

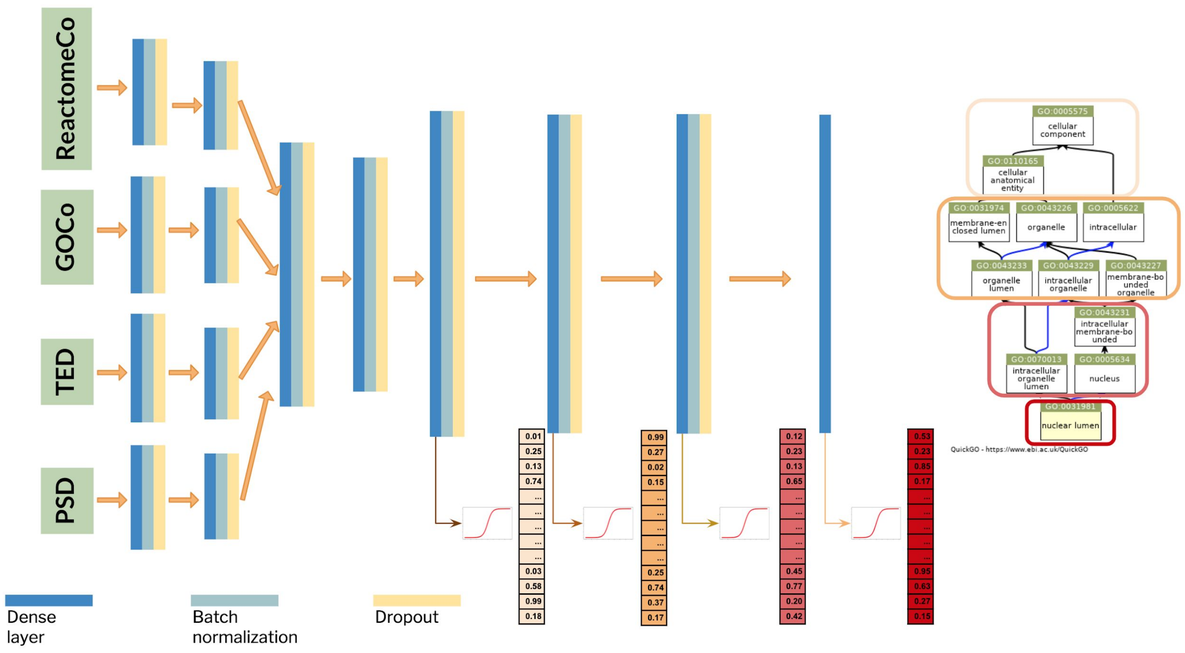

Figure 1: Deep learning model. For a particular GO sub-ontology (i.e BP, CC, or MF), the network could have different inputs from association matrices: proteins sequence distance (PSD); transcriptomics expression data (TED); co-occurrence in the other GO sub-ontologies (GOCo); and co-occurrence in Reactome pathways (ReactomeCO).

Experimental setup

- Grid search for hyper-parameters optimization with 3-fold cross validation (70%-10%-20%):
  - Probability of dropout: [**0.25**,0.5, 0.75].
  - Batch size: [8, **16**, 32, 64, 128].
  - Activation functions: [ReLU, **ELU**, CeLU, tanh].
  - Optimizer method=[**Adam**, Adadelta, RMSprop].
  - Loss criterion=[SmoothL1Loss, **BCELoss**, CrossEntropyLoss].
- Early stopping (patience =10) .
- Evaluation by performance measures used in CAFA challenges [2].

Results

- The lowest F-max was reached considering only PSD as input data: MF, 0.38; CC, 0.63; BP, 0.35.
- Integrating PSD with GOCo improves performance scores, mainly for MF and BP.
- With PSD, GOCo and ReactomeCo as input data, F-max was improved because of an increment in recall and precision for BP and CC, respectively.
- For MF and CC terms prediction, considering TED allowed increasing the F-max.
- The highest F-max (0.65) achieved using all input matrices for CC.

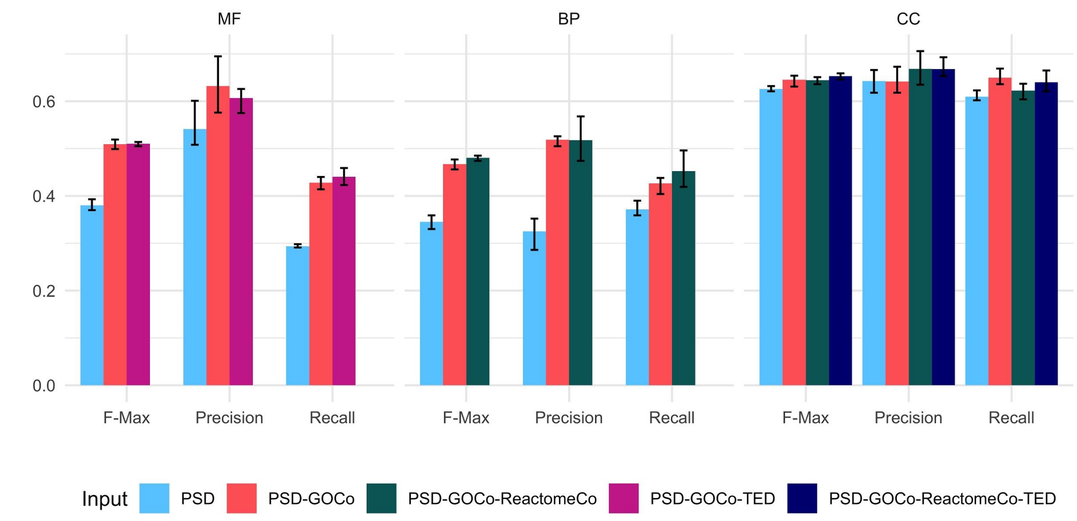

Figure 2: Model performances. Barplots depicting average scores reached by the proposed deep learning models feeded by different input data. Error bars indicate minimum and maximum values.

Conclusions

- Protein knowledge represented by several association matrices have shown to provide useful information for better function prediction.
- Integrating protein sequences with knowledge from multiple data sources improved model performances.
- As more input data were integrated, better F-max were reached.
- Deep learning models integrating multi-source data are a promising tool for protein function prediction.

References

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